Mortgage Delinquency Risk Analysis

Portfolio-grade credit risk analytics — Fannie Mae Single-Family Loan Performance Dataset (2020 Q1–Q2)

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Business Problem

Within a large mortgage portfolio, which borrowers are most likely to become delinquent, why, and what should management do about it? A single portfolio-wide delinquency rate hides the segments where risk and exposure actually concentrate — this project builds the analytical layer beneath that number.

Objectives

- Build a reliable portfolio health baseline
- Identify borrower, structural, and geographic delinquency drivers
- Quantify exposure at risk by segment, not just rate
- Test loan modification as an independent risk signal
- Convert findings into a monitoring framework with thresholds

Dataset Overview

- Fannie Mae Single-Family Loan Performance, 2020 Q1–Q2
- 84.18M loan-month records after cleaning
- Reduced to ~1.92M row latest-snapshot table for reporting
- 108 raw fields mapped to Fannie Mae glossary
- Full SQL pipeline: raw ingestion to reporting layer

Dashboard Preview

1. Executive Overview	How healthy is the portfolio overall?
2. Borrower Risk	Which borrowers are most likely to become delinquent?
3. Property & Loan	Which loan and property structures carry elevated risk?
4. Geographic Risk	Why are certain markets riskier than others?
5. Behavioural Risk	How does modification status affect delinquency?
6. Portfolio Surveillance	Which segments need continuous monitoring?

Key Findings

Modification is the strongest risk signal — Modified loans: 22.8% delinquent vs 0.6% non-modified (38x), independent of risk segment.

Credit score dominates individual risk — 22x delinquency spread between Very High Risk and Very Low Risk borrowers.

Risk layering compounds — Multi-factor risky borrowers delinquent at 2.4% vs 0.3% baseline — nearly 8x.

Geography follows borrower composition — New York elevated on borrower quality, not location; California carries 16.6% of exposure at just 0.47% delinquency.

Exposure concentration is not delinquency-proportional — Moderate Risk segment holds $1.3bn in delinquent exposure vs $0.5bn for High Risk — a volume effect.

Purchase loans run ~2x refinance delinquency — 1.1% vs 0.6%, likely a COVID-period origination timing effect.

Business Recommendations

1. Monitor modified loans as an independent segment, regardless of credit risk tier.
2. Apply stricter underwriting where multiple risk factors coincide simultaneously.
3. Treat geographic outliers as a borrower-composition question before adjusting state-level lending policy.
4. Build house price stress scenarios for high-exposure, low-delinquency states.
5. Extend to roll rate and cure rate analysis using the existing 84M-row panel.

Project Deliverables

Repository Navigation

- Executive Credit Risk Report
- Business Case Study
- Technical Documentation
- Six-page Power BI dashboard
- Full SQL pipeline (8 stages)
- Column mapping and data dictionary

Start with **README.md** for full detail.
Business-facing findings: **01_Project_Documentation/** Executive_Credit_Risk_Report.docx
Technical reviewers: **03_SQL/** and Technical_Documentation.docx
Dashboard screenshots: **02_Dashboard/**